The proposed project is an **AI-Based Intelligent Question Paper Generation System** that automatically generates customized and multiple unique question-paper sets based on an institute's official course plan. The system will provide a web-based **GUI with login/signup**, where faculty can enter the institute and subject details, upload the course plan in **PDF/DOC/DOCX format**, select the number of question-paper sets, choose the required modules, specify the **total marks**, and **define their own question and marks distribution**, such as the number of 2-mark, 5-mark, 10-mark, or other required question types. The faculty can also specify how many questions of each mark category should be generated from **each selected module** and, where required, select the desired **CO and cognitive-level distribution**. The system will not impose a fixed marks pattern; instead, it will generate questions according to the user's selected requirements and validate that the complete distribution matches the specified total marks. The uploaded course plan will be analyzed using **NLP, document processing, embeddings, and RAG (Retrieval-Augmented Generation)** to identify modules, topics, subtopics, and course content, Course Outcomes (COs), and their associated cognitive levels. Based on this information and the faculty's constraints, an LLM will generate **syllabus-oriented, conceptual, application-based, scenario-based, and higher-order questions** with appropriate difficulty and Bloom's Taxonomy levels. **Every generated question should display its corresponding CO level along with the marks**.

A major feature of the system will be **multi-set generation**, where every set contains unique questions while maintaining the same marks, module distribution, syllabus coverage, and comparable difficulty. **Semantic similarity techniques** will be used to detect and eliminate questions that are different in wording but essentially the same. A validation engine will check marks, module-wise distribution, syllabus relevance, duplicate questions, difficulty, and other constraints before finalizing the papers. Faculty members will be able to **preview, edit, delete, regenerate, and approve** questions before final submission. All users, course plans, questions, question banks, generated sets, and previous papers will be stored in a database(eg. Presidency university>DBMS>SetA,SetB), and approved papers can be exported as **PDF or DOCX**. Thus, the system aims to reduce faculty workload while producing **unique, balanced, syllabus-grounded, and customizable question papers automatically**.